

Engineering Physics (2025)

Course code 25PY101

Unit 2: Quantum mechanics

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Unit 2 Plan

- 1 Introduction to QM
- 2 Dual nature of radiation
- 3 de Broglie's concept of matter waves
- 4 Heisenberg's uncertainty principle
- 5 Schrödinger's time dependent wave equation
- 6 Particle in a 1D box

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Applications of Schrödinger's W.E.

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi(x) = 0 \quad \Psi(x, t) = \psi(x)e^{-j\omega t}$$

- The aim is to solve for the time independent wave function $\psi(x)$ for a given potential $V(x)$.
- The potential functions that are useful to describe commonly occurring physical systems are

$V(x)$	Name	Physical systems
0	Constant potential	Particle in free space
$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$	Infinite potential well	Ideal atom
$V(x) = \begin{cases} 0 & -\frac{a}{2} < x < \frac{a}{2} \\ V_0 & \text{otherwise} \end{cases}$	Finite potential well	Work function
$V(x) = \begin{cases} V_0 & -\frac{a}{2} < x < \frac{a}{2} \\ 0 & \text{otherwise} \end{cases}$	Finite barrier	Nuclear fission
		Scanning Tunneling Micro

Particle in free space

$V(x)$: Constant potential

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad \Psi(x, t) = \psi(x)e^{-j\omega t}$$

- Let us define the wave vector k as

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

so that the Time Independent Schrödinger Equation (TISE) is reduced to

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0$$

so that

$$\frac{d^2\psi(x)}{dx^2} = -k^2\psi(x)$$

$V(x)$: Constant potential – Solution of TISE

$$\frac{d^2\psi(x)}{dx^2} = -k^2\psi(x)$$

-
- The solution of above differential equation can be written as

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

- If we use the Euler's formula

$$e^{jx} = \cos x + j \sin x$$

then the solution can be written also as

$$\psi(x) = A' \cos kx + B' \sin kx$$

$V(x)$: Constant potential – Wave vector, Momentum and Energy

- All the wave vectors are allowed

$$k = -\infty \text{ to } +\infty$$

- The momentum is given by

$$p = \frac{h}{\lambda} \Rightarrow p = \hbar k \quad \left[\because k = \frac{2\pi}{\lambda} \right]$$

- The corresponding energy levels are given by

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

$V(x)$: Constant potential – Solution of TDSE

- The solution of Time Dependent part of Schrödinger Equation (TDSE) is given by

$$\phi(t) = e^{-j\omega t}, \quad \text{where} \quad \omega = \frac{E}{\hbar} = \frac{\hbar k^2}{2m}$$

- The total wave function is given by

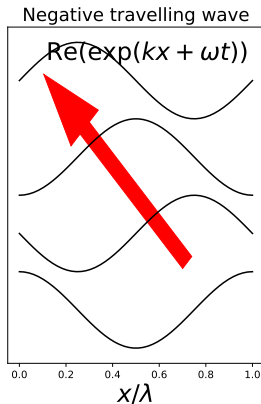
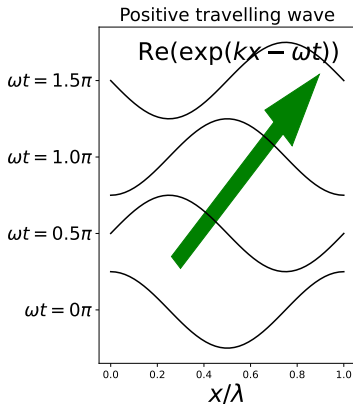
$$\Psi_k(x, t) = \psi_k(x)\phi_k(t) = e^{j\overline{kx - \omega t}}.$$

where the subscript k is the quantum number.

$V(x)$: Constant potential – Nature of solutions

- The solutions are **traveling** waves.
- The energy E can be any positive real number and is related to the wave vector k by

$$E = \frac{\hbar^2 k^2}{2m}$$



Constant potential: problems

Problem

An electron in free space is described by a plane wave given by $\Psi(x, t) = A \exp(j(kx - \omega t))$. If $k = 8 \times 10^8 \text{ m}^{-1}$ and $\omega = 8 \times 10^{12} \text{ rad s}^{-1}$, determine

- ① *phase velocity and wavelength of the plane wave*
- ② *momentum and kinetic energy (in eV)*

Repeat the steps for $k = -1.5 \times 10^9 \text{ m}^{-1}$ and $\omega = 1.5 \times 10^{13} \text{ rad s}^{-1}$.

Problem

Determine the wave number, wavelength, angular frequency, and period of wavefunction that describes an electron traveling in free space at a velocity of (a) $5 \times 10^6 \text{ cm s}^{-1}$.

Summary of particle in free space

- For a particle in free space, all wave vectors are allowed $k = -\infty$ to $+\infty$.
- The corresponding energy levels are given by

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

- The time independent part of wave function is given by

$$\psi(x) = e^{jkx}$$

- The time dependent part of wave function is given by

$$\phi(t) = e^{-j\omega t}, \quad \text{where} \quad \omega = \frac{E}{\hbar}$$

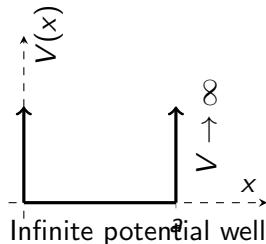
- The total wave function is given by

$$\Psi_k(x, t) = \psi_k(x)\phi_k(t) = e^{j\overline{kx - \omega t}}.$$

Particle in a 1D box

$V(x)$: Infinite potential

$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$$



- Consider a particle in a 1-dimensional infinite potential well.
- The well can be considered as a box.

Outside infinite box

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - \infty)\psi(x) = 0$$

Inside infinite box

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0$$

$V(x)$: Infinite potential well – Solution of TISE

Outside infinite box

- There is no wave function!

$$\psi(x) = 0$$

- Upon applying boundary conditions

Inside infinite box

- The wavefunction is

$$\psi(x) = A \cos kx + B \sin kx$$

Left B.C.

$$\psi(x=0) = 0$$

$$A \cos kx = 0$$

$$\Rightarrow A = 0$$

Right B.C.

$$\psi(x=a) = 0$$

$$B \sin ka = 0$$

$$\Rightarrow \sin ka = 0$$

$V(x)$: Infinite potential well – Solution of TISE

- The equation $\sin ka = 0$ is valid if

$$ka = n\pi$$

where n is a positive integer, or $n = 1, 2, 3, \dots$. The parameter is referred to as quantum number. In terms of the quantum number, the wave vector is given by

$$k = \frac{n\pi}{a}$$

- The prefactor B can be found by applying the **normality condition** of wave function.

$$\int_0^a |B \sin kx|^2 dx = 1, \quad \Rightarrow \quad \int_0^a B^2 \sin^2 kx dx = 1$$

$$\therefore B = \sqrt{\frac{2}{a}}$$

- Therefore, the time independent wave function is given by

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

$V(x)$: Infinite potential well – Energy levels

- Recall that the wave vector k is related to the energy E by

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad \Rightarrow \quad E = \frac{\hbar^2 k^2}{2m}$$

- Therefore, the energy is given by

$$E = E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}$$

Problem

Electron in an infinite potential well of width $5 \text{ \AA}$. Calculate the first three energy levels.

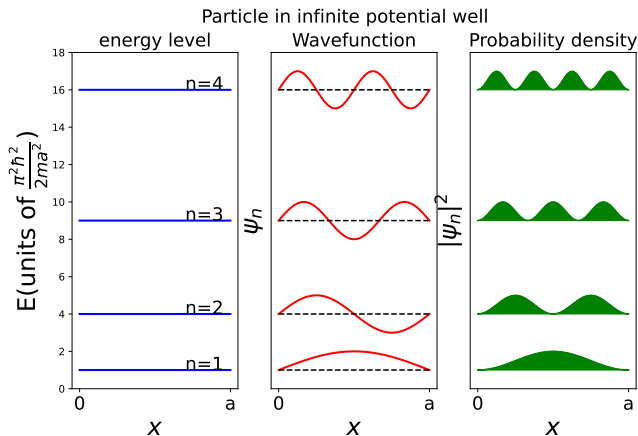
Key Insight

The energy of particle is **quantized** i.e. it can only have particular **discrete** values.



$V(x)$: Infinite potential well – Nature of solutions

- The solutions are **standing** waves.



Orthogonality condition

- Wave function associated with an energy level is normalized.
- In addition to normality, wave functions $\psi_m(x)$, $\psi_n(x)$ corresponding to any two levels m, n of infinite potential well are orthogonal to each other i.e.

$$\int_0^a \psi_m^*(x) \psi_n(x) dx = 0.$$

This is called the **orthogonality condition**.

Theorem

Prove the orthogonality condition for solutions of infinite potential well.

1D infinite well: problems

Problem

- 1 A particle with mass of 15 mg is bound in a 1D infinite potential well that is 1.2 cm wide.
- 2 If the energy of the particle is 15 mJ, determine the value of n for that state.
- 3 What is the energy of the $(n + 1)$ state?
- 4 Would quantum effects be observable for this particle?

Summary of particle in infinite potential well

- For a particle in an infinite potential well, discrete wave vectors are allowed and are given by

$$k_n = \frac{n\pi}{a}.$$

- The corresponding energy levels are given by

$$E_n = \frac{\hbar^2 k^2}{2m}, \quad \text{or} \quad E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}$$

- The time independent part of wave function is given by

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

- The time dependent part of wave function is given by

$$\phi_n(t) = e^{-j\omega_n t}, \quad \text{where} \quad \omega_n = \frac{E_n}{\hbar}$$

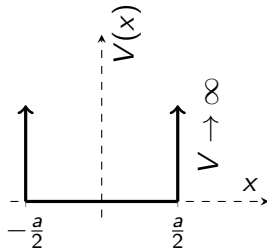
- The total wave function is given by

$$\Psi_n(x, t) = \psi_n(x)\phi_n(t),$$

Particle in a symmetric form infinite potential well

$V(x)$: Symmetric Infinite potential

$$V(x) = \begin{cases} 0 & -\frac{a}{2} \leq x \leq +\frac{a}{2} \\ \infty & \text{otherwise} \end{cases}$$



- Consider a particle in a **symmetric form** of 1-dimensional infinite potential well.
- The analysis for this potential is similar to the analysis for the asymmetric form discussed earlier.
- However, the general form of wave function can be considered as linear combination of exponential functions instead of trigonometric functions due to the symmetry of the problem.
- Thus, let the wave function inside the well be

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

$V(x)$: Symmetric Infinite potential – Solution of TISE

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

- Upon applying left boundary condition,

$$\psi(x = -\frac{a}{2}) = 0$$

$$\Leftrightarrow A \exp\left(-j\frac{ka}{2}\right) + B \exp\left(+j\frac{ka}{2}\right) = 0$$

$$\Leftrightarrow (A + B) \cos\left(\frac{ka}{2}\right) + j(-A + B) \sin\left(\frac{ka}{2}\right) = 0$$

$$\Leftrightarrow (A + B) \cos\left(\frac{ka}{2}\right) = 0 \text{ and } (-A + B) \sin\left(\frac{ka}{2}\right) = 0$$

$$\Leftrightarrow [A + B = 0 \text{ or } \cos\left(\frac{ka}{2}\right) = 0] \text{ and } [A - B = 0 \text{ or } \sin\left(\frac{ka}{2}\right) = 0]$$

$$\Leftrightarrow [A - B = 0 \text{ and } \cos\left(\frac{ka}{2}\right) = 0] \text{ or } [A + B = 0 \text{ and } \sin\left(\frac{ka}{2}\right) = 0]$$

- Same cases arise for the right boundary condition due to **symmetry** of the problem!

$V(x)$: Symmetric Infinite potential – Solution of TISE

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

Case (i) Even wave functions

$$A - B = 0$$

and

$$\cos\left(\frac{ka}{2}\right) = 0$$

$$\Leftrightarrow \psi(x) = A \cos kx$$

$$\Leftrightarrow \frac{ka}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$\Leftrightarrow k = \frac{\pi}{a}, \frac{3\pi}{a}, \dots$$

Case (ii) Odd wave functions

$$A + B = 0$$

and

$$\sin\left(\frac{ka}{2}\right) = 0$$

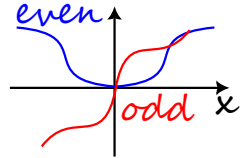
$$\Leftrightarrow \psi(x) = A \sin kx$$

$$\Leftrightarrow \frac{ka}{2} = \pi, 2\pi, \dots$$

$$\Leftrightarrow k = \frac{2\pi}{a}, \frac{4\pi}{a}, \dots$$

$V(x)$: Symmetric Infinite potential – Nature of solutions

$$f(x) := \begin{cases} \text{even} & \text{if } f(-x) = f(x) \\ \text{odd} & \text{if } f(-x) = -f(x) \end{cases}$$



- The wave functions belonging to first case are even functions as $\cos x$ is an even function whereas The wave functions belonging to second case are odd functions as $\sin x$ is an odd function.
- There are two subfamilies of even and odd wave functions
- If we order the wave functions with increasing wave vector, then wave functions alternate between even and odd.

$$\psi_e^{(2)}(x) \propto \sin\left(\frac{4\pi x}{a}\right)$$

$$\psi_e^{(2)}(x) \propto \cos\left(\frac{3\pi x}{a}\right)$$

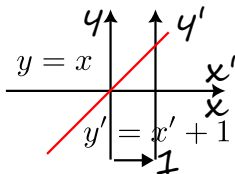
$$\psi_o^{(1)}(x) \propto \sin\left(\frac{2\pi x}{a}\right)$$

$$\psi_e^{(1)}(x) \propto \cos\left(\frac{\pi x}{a}\right)$$

$V(x)$: Symmetric Infinite potential – Alternative solution

- Translation of origin by α transforms the function in new coordinate frame

$$f^{\text{new}}(x) = f^{\text{old}}(x + \alpha)$$



- The solutions to the symmetric form of infinite potential well can be obtained from the solutions to the asymmetric form by translating the coordinate frame by $\frac{a}{2}$.

$$x \rightarrow x + \frac{a}{2}$$

- The wave functions of the symmetric potential well are given by

$$\psi^{\text{sym}}(x) = \psi^{\text{asym}}\left(x + \frac{a}{2}\right)$$

$$\psi^{\text{sym}}(x) = \psi^{\text{asym}}\left(x + \frac{a}{2}\right)$$

$$= \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x + \frac{n\pi}{2}\right) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a} + \frac{n\pi}{2}\right)$$

$V(x)$: Symmetric Infinite potential – Alternative solution

- Upon substituting for $n = 1, 2, \dots$, we have

$$\psi^{\text{sym}}(x) \propto \begin{cases} \sin\left(\frac{\pi x}{a} + \frac{\pi}{2}\right) & n = 1 \\ \sin\left(\frac{2\pi x}{a} + \frac{2\pi}{2}\right) & n = 2 \\ \sin\left(\frac{3\pi x}{a} + \frac{3\pi}{2}\right) & n = 3 \\ \sin\left(\frac{4\pi x}{a} + \frac{4\pi}{2}\right) & n = 4 \\ \vdots & \vdots \end{cases}$$

- Using the trigonometric relations, we have

$$\psi^{\text{sym}}(x) \propto \begin{cases} \cos\left(\frac{\pi x}{a}\right) & n = 1 \\ \sin\left(\frac{2\pi x}{a}\right) & n = 2 \\ \cos\left(\frac{3\pi x}{a}\right) & n = 3 \\ \sin\left(\frac{4\pi x}{a}\right) & n = 4 \\ \vdots & \vdots \end{cases}$$

In some cases, prefactor of -1 can be absorbed into normalizing factor.

- Thus, the solutions are grouped into even and odd families.

Summary of particle in symmetric infinite potential well

- For a particle in a symmetric infinite potential well, the allowed wave vectors are same as that of asymmetric infinite potential well.
- The corresponding energy levels are also same
- The potential well function is an even function i.e.

$$V(-x) = V(x).$$

Due to the symmetry of the potential well function, the wave functions are either **odd** or **even**.